

	Value Driver	Current State (Baseline)	Target State	Key Assumption	Illustrative Impact
	Deviation investigation effort	Avg. 8–12 hrs per deviation; manual root cause analysis	4–6 hrs per deviation with guided workflows	50% of deviations are recurring or process-related	40–50% reduction in investigation time
	Batch execution & review cycle time	5–7 days from batch start to release	3–4 days with real-time review enablement	Review delays driven by paper-based handoffs	30–40% reduction in cycle time
	Manual documentation effort	Significant operator time spent on manual documentation	20–30% via automated data capture	Core process steps are MES-integrated	50–60% reduction in manual documentation
	Right-first-time execution	85–90% RFT rate across batch operations	Improved RFT through enforced sequencing + controls	Errors are largely procedural, not equipment-driven	5–8 percentage point improvement in RFT
	Manufacturing visibility & decision support	Lagging KPIs; weekly/monthly reporting cycles	Near real-time dashboards and exception alerts	Key data sources integrated into MES layer	Shift from reactive to proactive decision-making
Impacts are illustrative and directional. Actual benefits depend on scope, site maturity, adoption, validation constraints, and implementation approach.					